

## "Vehicle Selector Pro" (Ruby on Rails)

This requirement is designed to build production-grade Shopify applications using **Ruby on Rails**. You will build an advanced version of a vehicle selector app that lets merchants assign vehicle attributes (Year, Make, Model, Trim, Engine, etc.) to products and provides a storefront widget for customers to filter products dynamically. The solution must demonstrate deep knowledge of Rails architecture, ActiveRecord, background processing, caching, security, testing, and integration with Shopify's APIs and app ecosystem.

This is **not a quick prototype**. You have ample time to deliver a polished, well-documented, and thoroughly tested solution that would be ready for submission to the Shopify App Store.

### Core Project Scope

Your task is to build a custom Shopify app that allows merchants to assign vehicle attributes (Year, Make, Model, etc.) to products and displays a dynamic filtering widget on the storefront.

- **Ruby on Rails Architecture:** The entire app must be a Rails engine/application utilizing the latest stable version (7.x+).
- **Authentication & Multitenancy:** Implement the standard OAuth flow using the `shopify_app` gem, ensuring secure and isolated data storage per shop using explicit scoping.
- **Admin UI:** Build a clean, responsive dashboard using Rails views (ERB/Haml) styled with Polaris CSS.
- **Data Architecture:** Store attribute data using Shopify Metafields. You must also maintain a normalized local PostgreSQL cache using ActiveRecord models for lightning-fast querying.
- **Storefront Extension:** Create a Shopify Theme App Extension to render the filter dropdowns natively within the theme editor (do not use the deprecated ScriptTag API).
- **App Proxy Server:** Build robust Rails controllers for your App Proxy endpoints to dynamically serve available filter options and return matching product IDs.
- **GraphQL & Webhooks:** Utilize the GraphQL Admin API for syncing data. Process all webhooks asynchronously using ActiveJob and Sidekiq.

### Expected Deliverables

Please provide a Git repository containing your Rails application and Theme App Extension. Include a straightforward README with setup instructions and a brief screen recording (2–3 minutes) demonstrating the working app.